MODULE 17 · DOMAIN 4 - EVALUATION, TESTING AND OPTIMISATION

Metrics and Success Criteria

Writing success criteria that can be tested, metrics for every dimension, the proxies that mislead, and matching each metric to its requirement.

DOMAIN 4 of 7 EXAM WEIGHT 16% MODULE 17 of 38

admin@sthithapragna.com

Domain 4 at a glance

OBJECTIVE	MODULE
Metrics for accuracy, latency, cost, safety, security	17
Eval datasets and test frameworks, mixed methods	18
A/B testing and iteration	19
Diagnosing prompt failure, hallucination, model mismatch	20
Cost and latency optimisation; monitoring in production	21

ON THE EXAM

One discipline runs through the domain: decide what good means before you build, change or ship.

A metric is a promise about what matters. **Measure the wrong thing, and you will improve the wrong thing.**

Good success criteria, in Anthropic's terms

- **Specific:** not good performance, but accurate sentiment classification.
- **Measurable:** numbers, or well-defined scales. Even safety can be counted.
- **Achievable:** based on baselines, benchmarks or experiments, within today's models.
- **Relevant:** tied to the purpose. Citation accuracy matters more in medicine than in chat.
- **Multidimensional:** most use cases need several criteria at once.

> ON THE EXAM
>
> Bad: safe outputs. Good: under 0.1% of 10,000 outputs flagged by the content filter.

From a wish to a criterion

A WISH

Quillon's first draft

```
The portal assistant
should be accurate,
fast and safe.
```

Nothing to test, nothing to fail, nothing to sign off.

CRITERIA

Rewritten

```
On 600 held-out portal questions:
  92% correct against the rubric
  p95 first token under 2 s
  no clinical advice on 200
    adversarial prompts
  under USD 0.04 per conversation
```

Specific, measurable, achievable, relevant. Four dimensions.

THE TIEBREAKER

Every line names a measure, a threshold, and the data it is measured on.

A metric for every dimension

DIMENSION	TYPICAL METRIC
Accuracy	Correct against a rubric; F1 for classes; claims grounded in sources
Latency	p50 and p95 first token and end-to-end, over a stated window
Cost	Cost per request, and per successful outcome
Safety	Policy violation rate on an adversarial set
Security	Access-control and injection results on a red-team set
Consistency	Agreement across repeated runs of the same task

ON THE EXAM

Cost per successful outcome beats cost per request: a cheap wrong answer is not cheap.

Proxy metrics that mislead

PROXY	WHY IT MISLEADS	MEASURE INSTEAD
Acceptance rate	People accept without checking	Expert-graded correctness on a sample
Text overlap with a reference	Same meaning, different words	Rubric or model grading of meaning
Average latency	Hides the slow tail	p95 and p99
Tickets closed	Rises while customers come back	Confirmed resolution, repeat contacts
Thumbs-up rate	Few vote, and not at random	Sampled review

ON THE EXAM

If a number can rise while users are worse off, it is a proxy. Pair it with a direct measure.

System metrics and outcome metrics

SYSTEM METRICS

Is it working?

- Correct against the rubric
- p95 latency per step
- Cost per request
- Violation rate on the adversarial set

OUTCOME METRICS

Does it matter?

- Cases resolved, and staying resolved
- Complaints and repeat contacts
- Time saved that is actually redeployed
- Cost per successful outcome

THE TIEBREAKER

Report both layers. Without outcomes you cannot show value; without system metrics you cannot explain it.

Percentiles, tails and consistency

- **p50** is the typical case; **p95 and p99** are the tail users remember.
- **Every target names** a percentile, a window, and where it is measured.
- **pass@k:** at least one of k attempts succeeds. Suits tools where retrying is cheap.
- **pass^k:** all k attempts succeed. Suits agents customers rely on every time.
- **75% per attempt** gives only about 42% for three in a row.

> THE PICTURE
>
> A striker needs one goal in three attempts. A penalty taker in a shoot-out must score every time.

Matching a metric to a requirement

01

Read it

What does the requirement actually promise?

02

Name the dimension

Accuracy, latency, cost, safety, security.

03

Pick the measure

With percentile, window and unit.

04

Set the threshold

From a baseline, not a hope.

05

Choose the data

A set that represents real use.

ON THE EXAM

Define the metrics and thresholds before you instrument or build. Measure what you promised.

> Mereside's agent-assist pilot reports that advisers accepted 96% of Claude's suggestions during calls.
>
> Over the same period, customer complaints about wrong information given on calls rose.

Which metric should become the primary quality measure?

A Acceptance rate, with the target raised to near total acceptance

B Correctness of suggestions, graded by policy experts on a regular sample of real calls

C Average call handling time

D Number of suggestions shown per call

◆ PAUSE AND DECIDE

Which metric should become the primary quality measure?

A Acceptance rate, with the target raised to near total acceptance

Rewards trust, not truth. The complaints show the gap.

B Correctness of suggestions, graded by policy experts on a regular sample of real calls

Measures what the complaints are about.

C Average call handling time

Efficiency, not accuracy. Wrong answers can be fast.

D Number of suggestions shown per call

Volume, not quality.

THE SIGNAL

Acceptance rising while errors rise: acceptance is a proxy. Measure correctness directly.

> Quillon's draft success criterion for prior-authorisation letters reads: letters should be high quality and produced quickly.
>
> Clinicians approve every letter. Insurers decide independently.

Which rewrite is the best success criterion?

A. Letters should be excellent and quick, as judged by the project team

B. Every letter is approved by the insurer on first submission

C. On held-out real cases: nine in ten letters need no clinical correction, every claim cites the record, and ninety-five in a hundred drafts arrive within a minute

D. Letters score highly on text overlap with past approved letters

◆ PAUSE AND DECIDE

Which rewrite is the best success criterion?

A Letters should be excellent and quick, as judged by the project team

Still a wish, judged by the people being judged.

B Every letter is approved by the insurer on first submission

Not achievable, and outside the system's control.

C On held-out real cases: nine in ten letters need no clinical correction, every claim cites the record, and ninety-five in a hundred drafts arrive within a minute

Specific, measurable, achievable, relevant, and multidimensional.

D Letters score highly on text overlap with past approved letters

A proxy. Each case differs; overlap is not correctness.

THE SIGNAL

Pick the criterion with a measure, a threshold and a dataset for each dimension that matters.

> Mereside's staff policy assistant now filters retrieval by each user's entitlements. The security requirement says staff without HR access must never see HR-restricted content.
>
> Corvane must add a metric for this requirement.

Which metric best measures it?

A The number of complaints from staff about leaked content

B On a red-team set of queries from users without HR access, the rate at which restricted content appears, with a target of zero

C Overall answer accuracy across all staff questions

D The model's own report of whether it used restricted documents

◆ PAUSE AND DECIDE

Which metric best measures it?

A. The number of complaints from staff about leaked content
Counts only leaks someone noticed and reported.

B. On a red-team set of queries from users without HR access, the rate at which restricted content appears, with a target of zero
Tests the control directly, with deliberate attempts.

C. Overall answer accuracy across all staff questions
A quality metric. Says nothing about access.

D. The model's own report of whether it used restricted documents
Self-report. The control lives outside the model.

THE SIGNAL

Security requirements are measured by deliberate attempts to break them, on a red-team set.

> Tallyhawk's support contract promises replies within 3 seconds. The dashboard shows an average of 1.9 seconds.
>
> Enterprise customers still complain that replies are often slow.

What should Corvane change first?

A Nothing, because the average is well within the promise

B Move the support agent to a faster model tier

C Report the slowest replies, at the ninety-fifth and ninety-ninth percentiles, against the promise

D Report the median reply time instead of the average

◆ PAUSE AND DECIDE

What should Corvane change first?

A Nothing, because the average is well within the promise

Averages hide exactly the slow replies customers describe.

B Move the support agent to a faster model tier

A fix before the measurement shows what is slow.

C Report the slowest replies, at the ninety-fifth and ninety-ninth percentiles, against the promise

Measures the tail the complaints are about.

D Report the median reply time instead of the average

Still the typical case. The tail stays hidden.

THE SIGNAL

Complaints about slowness with a healthy average: measure the tail before changing anything.

> Tallyhawk's billing agent adjusts invoices after plan changes. Customers expect it to work every time they ask.
>
> Product wants to report it as 95% successful, because on each test task at least one of three attempts succeeded.

What should Corvane recommend?

A. Measure success on every attempt, such as all of three trials passing

B. Keep the at-least-one-of-three measure, as proposed

C. Report the best single run from the test week

D. Report the share of tasks any recent model version has ever solved

◆ PAUSE AND DECIDE

What should Corvane recommend?

A. Measure success on every attempt, such as all of three trials passing
Matches the promise: right every time.

B. Keep the at-least-one-of-three measure, as proposed
Customers do not get three tries at an invoice.

C. Report the best single run from the test week
Cherry-picking. Says nothing about reliability.

D. Report the share of tasks any recent model version has ever solved
Not this system, not this version.

THE SIGNAL

Users rely on it every time: pass^k. Retries are cheap and someone checks: pass@k.

> Mereside's complaints triage must classify each complaint into the regulator's categories, including rare ones.
>
> It must also never miss an indicator that a customer may be vulnerable. Corvane is choosing the metrics for sign-off.

Which two metrics most directly measure these requirements? Choose two.

A Recall on vulnerability indicators, on an expert-labelled set

B Average length of the drafted responses

C Handlers' acceptance rate for the drafts

D Per-category accuracy, including rare categories, on an expert-labelled set

E Text overlap between drafts and past responses

◆ PAUSE AND DECIDE

Which two metrics most directly measure these requirements? Choose two.

A Recall on vulnerability indicators, on an expert-labelled set

Never miss means recall: the share of real cases caught.

B Average length of the drafted responses

Unrelated to either requirement.

C Handlers' acceptance rate for the drafts

A proxy, and about drafts, not classification.

D Per-category accuracy, including rare categories, on an expert-labelled set

An overall score would hide failures on rare categories.

E Text overlap between drafts and past responses

A proxy for draft wording, not triage.

THE SIGNAL

Never miss means recall. Rare categories mean per-category results, not one overall number.

In five lines

- **Criteria:** specific, measurable, achievable, relevant, and multidimensional.
- **Every dimension gets a metric:** accuracy, latency, cost, safety, security, consistency.
- **Beware proxies:** acceptance, overlap, averages, tickets closed, thumbs up.
- **Tails and reliability:** p95 and p99; pass^k when users rely on every attempt.
- **Match the metric to the promise,** and set it before you build.

COMING UP

Next: building eval datasets and test frameworks.